

1. Problem Statement: Predict credit default risk.

2. Architecture: Streamlit UI, LightGBM ML, LangChain NL-to-SQL.

3. Dataset Overview: 300k+ rows, 122 features (application_train).

4. EDA Insights: Low income & missed payments correlate to risk.

5. ML Pipeline: Preprocessing -> SMOTE -> LightGBM.

6. Evaluation Metrics: ROC-AUC, F1, Precision, Recall.

7. SHAP Explainability: Feature importance & local explanations.

8. Business Rules: IF income < X AND EXT_SOURCE < Y THEN HIGH RISK.

9. NL-to-SQL Chatbot: Groq API querying SQLite DB securely.

10. Docker Deployment: docker-compose setup.

11. Challenges & Improvements: Handled class imbalance via SMOTE.

12. Conclusion: Production-ready credit risk platform deployed.